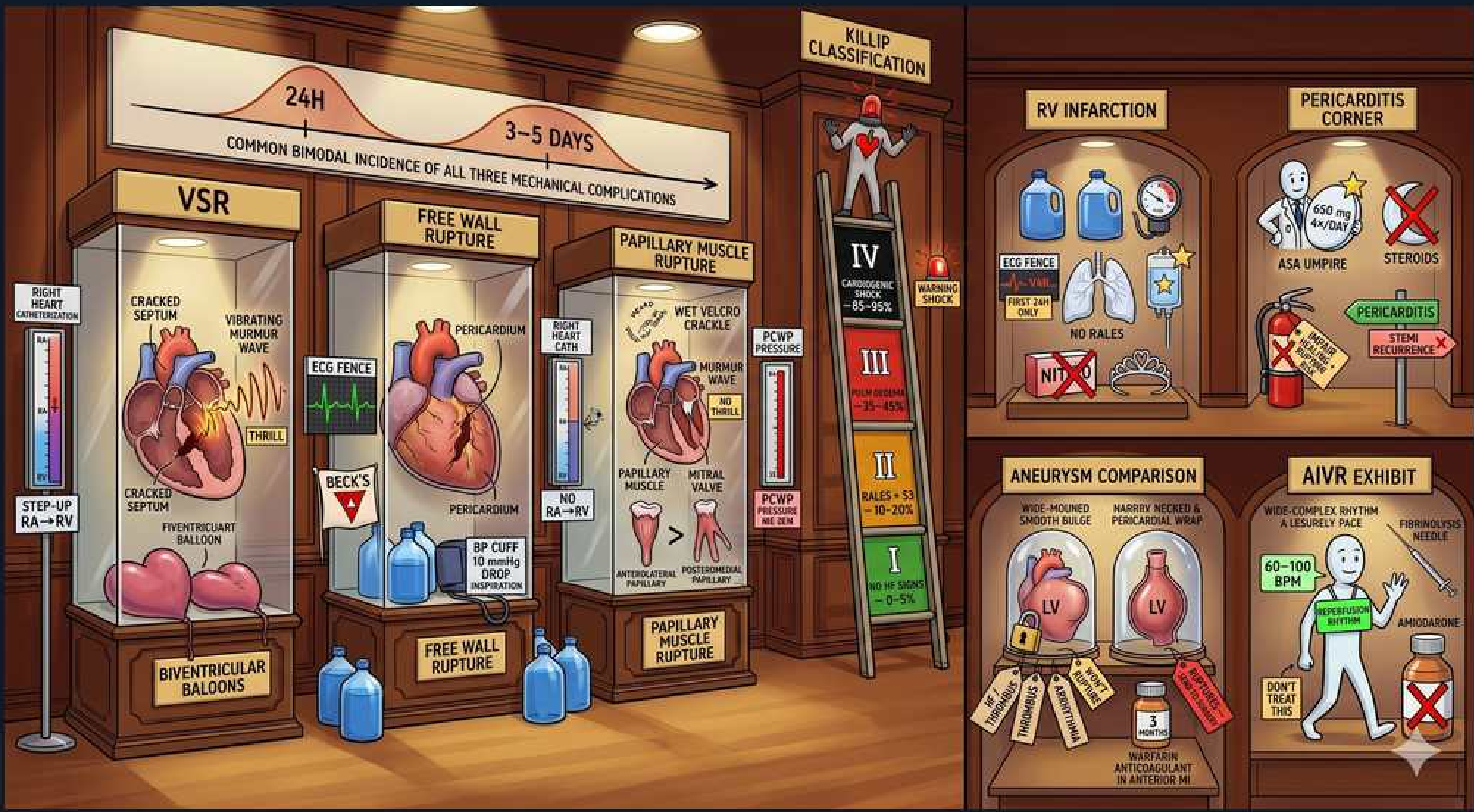
IHD / STEMI — mechanical complications, RV & pericarditis



1. Bimodal 24h / 3–5 days timing

WHAT YOU SEE

Top banner '24H' and '3–5 DAYS' bimodal

WHAT IT ENCODES

Mechanical (rupture) complications of STEMI have a bimodal incidence: an early peak within 24h (before much remodeling) and a second peak at days 3–5, when neutrophil-mediated removal of necrotic myocytes leaves the wall maximally softened and friable. Reperfusion (PCI/thrombolysis) reduces overall rupture rates but thrombolytics given late can paradoxically increase early rupture. The three rupture syndromes — VSR, free wall rupture, papillary muscle rupture — all cluster in this window. Mnemonic: 'rupture when the wall is mush: days 3–5.'

BOARD TRAP

Board says: 'Mechanical complications occur only weeks after the MI.' WRONG — that timing fits Dressler syndrome and LV aneurysm, not rupture. Discriminator: a NEW murmur or sudden hemodynamic collapse 3–5 days post-MI is a mechanical rupture, not a late autoimmune phenomenon.

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2. VSR cabinet

WHAT YOU SEE

Left glass museum cabinet labelled 'VSR' holding a heart with a visibly CRACKED septum and a 'vibrating murmur wave' — the split-open septum literally is the ventricular septal rupture creating a left-to-right shunt.

WHAT IT ENCODES

Ventricular septal rupture (VSR/VSD) creates a left-to-right shunt and presents as a NEW harsh holosystolic murmur at the left lower sternal border, often with a palpable thrill, plus acute biventricular failure and cardiogenic shock. It complicates anterior (apical septum, LAD) and inferior (basal septum, RCA) MIs; the latter carries worse prognosis. Diagnose with echo + oxygen step-up on right-heart cath. Definitive treatment is surgical repair (often after stabilization with IABP, vasodilators, inotropes). Mnemonic: 'septal rupture = shunt + thrill.'

BOARD TRAP

Board says: 'VSR causes no murmur, so it can be excluded by a clear exam.' WRONG. Discriminator: VSR characteristically produces a NEW loud holosystolic murmur with a thrill — its absence makes free wall rupture (no murmur, tamponade) more likely instead.

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3. VSR step-up RA→RV

WHAT YOU SEE

'STEP-UP RA→RV' label, biventricular balloons

WHAT IT ENCODES

The hallmark cath finding of VSR is an oxygen saturation STEP-UP from the right atrium to the right ventricle: oxygenated LV blood crosses the septal defect into the RV, raising RV/PA O2 saturation above RA. A step-up of ≥7% (e.g., RA 65% → RV 80%) confirms a left-to-right shunt at the ventricular level. This distinguishes VSR from acute MR (giant PCWP v-wave, no step-up). Mnemonic: 'shunt in the septum → O2 steps up.'

BOARD TRAP

Board says: 'VSR shows an oxygen step-DOWN from RA to RV.' WRONG. Discriminator: a left-to-right shunt adds oxygenated blood to the RV, so saturation STEPS UP (RA→RV). A step-down would imply a right-to-left shunt — not the VSR physiology.

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4. Free wall rupture pericardium

WHAT YOU SEE

Middle cabinet 'FREE WALL RUPTURE' showing a heart bleeding into the surrounding pericardium sac — the wall has torn open and blood floods the pericardium, the visual of tamponade.

WHAT IT ENCODES

Free wall rupture tears the LV myocardium into the pericardial space, causing acute hemopericardium → cardiac tamponade → pulseless electrical activity (PEA) and sudden death, typically days 3–5. Risk factors: first MI, anterior location, older age, female sex, late or no reperfusion, hypertension. A contained rupture forms a pseudoaneurysm. Emergent pericardiocentesis buys time; definitive treatment is immediate surgery. Mnemonic: 'free wall = free bleeding into the pericardium.'

BOARD TRAP

Board says: 'Free wall rupture presents with a loud new murmur.' WRONG — that's VSR or papillary muscle rupture. Discriminator: free wall rupture has NO murmur; it presents with sudden hypotension, JVD, muffled sounds (Beck's triad), and PEA arrest from tamponade.

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5. Beck's triad / tamponade

WHAT YOU SEE

'BECK'S' label, no RA→RV, BP cuff

WHAT IT ENCODES

Tamponade from free wall rupture produces Beck's triad: hypotension, jugular venous distension, and muffled (distant) heart sounds, plus pulsus paradoxus and often a rapid progression to PEA. There is NO oxygen step-up (no shunt) and NO holosystolic murmur. Bedside echo shows pericardial effusion with RV diastolic collapse. Mnemonic: 'Beck's three Ds: Down BP, Distended neck veins, Distant heart sounds.'

BOARD TRAP

Board says: 'Free wall rupture presents with an RA→RV oxygen step-up.' WRONG — that is VSR. Discriminator: free wall rupture bleeds into the pericardium (tamponade physiology, Beck's triad), creating no intracardiac shunt, so there is NO O2 step-up.

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6. Papillary muscle rupture

WHAT YOU SEE

Right cabinet 'PAPILLARY MUSCLE RUPTURE' with a snapped papillary muscle beside a 'wet velcro crackle' (flash pulmonary edema) — the torn muscle frees the mitral leaflet into acute severe MR.

WHAT IT ENCODES

Papillary muscle rupture causes acute SEVERE mitral regurgitation with flash pulmonary edema and cardiogenic shock, typically days 3–5. The POSTEROMEDIAL papillary muscle is most often involved because it has a single (RCA-dependent) blood supply, so inferior MIs are the classic setup. The murmur may be soft or absent because the regurgitant volume equalizes LA/LV pressure quickly. Treat with afterload reduction (nitroprusside), IABP, and urgent surgical repair/replacement. Mnemonic: 'posteromedial papillary has a poor single supply → inferior MI rupture.'

BOARD TRAP

Board says: 'Papillary muscle rupture causes a septal shunt with an O2 step-up.' WRONG — that's VSR. Discriminator: papillary muscle rupture causes acute MR (giant PCWP v-wave, pulmonary edema), NOT an intracardiac shunt — there is no oxygen step-up.

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7. Acute MR / PCWP v-wave

WHAT YOU SEE

'MITRAL VALVE', 'PCWP PRESSURE' giant v-wave

WHAT IT ENCODES

Acute MR from papillary muscle rupture produces a giant 'v wave' on the pulmonary capillary wedge pressure (PCWP) tracing, because the non-compliant left atrium receives a sudden large regurgitant volume in systole. There is NO oxygen step-up (the regurgitation is into the LA, not a left-to-right ventricular shunt). Echo (often TEE) confirms a flail leaflet. Mnemonic: 'acute MR → giant v-wave, no step-up.'

BOARD TRAP

Board says: 'Acute MR shows an RA→RV oxygen step-up on cath.' WRONG — that defines VSR. Discriminator: acute MR's signature is a giant PCWP v-wave with normal RA/RV saturations; a step-up means VSR, not MR.

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8. Killip classification ladder

WHAT YOU SEE

Tall ladder labelled 'KILLIP CLASSIFICATION' climbing rungs I→IV (each rung worse: clear lungs at the bottom up to 'CARDIOGENIC SHOCK 85–95%') with a red warning-shock light at top — climbing the ladder = climbing mortality.

WHAT IT ENCODES

The Killip classification grades acute heart-failure severity in STEMI and predicts mortality: Class I = no failure (no rales, no S3); Class II = rales <50% lung fields and/or S3; Class III = frank pulmonary edema (rales >50%); Class IV = cardiogenic shock (SBP <90, hypoperfusion), with ~80–95% historical mortality (lower with modern revascularization). Higher class = worse prognosis. Mnemonic: 'Killip climbs with congestion: I clear, IV shock.'

BOARD TRAP

Board says: 'The Killip class reflects infarct SIZE measured by biomarkers.' WRONG. Discriminator: Killip is a CLINICAL bedside grade of heart-failure/perfusion status, not a measure of infarct size or troponin level (though it correlates with prognosis).

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9. RV infarction — no rales

WHAT YOU SEE

Right shelf 'RV INFARCTION', lungs 'NO RALES'

WHAT IT ENCODES

RV infarction accompanies ~30–50% of inferior (RCA) MIs and presents with the triad of hypotension, elevated JVP (Kussmaul sign), and CLEAR lung fields (no rales) — because the failing RV cannot deliver enough preload to the left heart, so the lungs stay dry. Diagnose with right-sided leads (ST elevation in V4R). Mnemonic: 'RV infarct = high neck veins, clear lungs, low pressure.'

BOARD TRAP

Board says: 'RV infarct causes pulmonary edema with diffuse rales.' WRONG — left-sided failure causes rales. Discriminator: RV infarction gives hypotension + JVD with CLEAR lungs; finding clear lungs in a hypotensive inferior-MI patient points to RV infarct, not LV failure.

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10. RV infarct — no nitrates

WHAT YOU SEE

Fire extinguisher 'NO NIT' (no nitrates) RV shelf

WHAT IT ENCODES

The RV-infarcted ventricle is exquisitely PRELOAD-dependent, so treatment is IV fluid loading (normal saline boluses) to maintain RV filling, plus inotropes (dobutamine) if still hypotensive. AVOID preload-reducing drugs — nitrates, morphine, and diuretics — which can cause profound hypotension. Mnemonic: 'RV infarct: fill the tank, never drop the preload (no nitrates).'

BOARD TRAP

Board says: 'Treat the hypotension of an RV infarct with nitroglycerin.' WRONG and dangerous. Discriminator: nitrates drop preload and will crash an RV-infarct patient; the correct answer is IV fluids first (and avoid nitrates/morphine/diuretics).

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11. Pericarditis corner

WHAT YOU SEE

Right-side shelf 'PERICARDITIS CORNER' display grouping the post-MI pericarditis items (ASA umpire, fire-extinguisher, STEMI-recurrence sign) — the corner collects everything about pericarditis after MI.

WHAT IT ENCODES

Post-MI pericarditis comes in two forms separated by timing: (1) early peri-infarction pericarditis (peri-infarction/'pericarditis epistenocardica'), 1–4 days post-MI from local inflammation over a transmural infarct, with pleuritic chest pain and a friction rub; and (2) Dressler syndrome, an autoimmune pericarditis 2–10 weeks later with fever, pleuropericarditis, elevated ESR, and effusion. Mnemonic: 'early = neighbor inflammation (days); Dressler = delayed immune (weeks).'

BOARD TRAP

Board says: 'Pericarditis after an MI is always immediate (within hours).' WRONG. Discriminator: there are TWO peaks — early peri-infarction pericarditis (days) and Dressler syndrome (2–10 weeks, autoimmune). A pericardial friction rub appearing weeks later is Dressler, not the early form.

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12. ASA umpire, NO steroids/NSAIDs

WHAT YOU SEE

'ASA UMPIRE', steroids crossed out

WHAT IT ENCODES

Treat post-MI pericarditis with high-dose ASPIRIN (e.g., 650 mg q6–8h), often with colchicine. AVOID corticosteroids and non-aspirin NSAIDs (indomethacin, ibuprofen) in the early post-MI period because they impair infarct healing/scar formation and raise the risk of free wall rupture and ventricular remodeling. Aspirin is preferred because the patient already needs it for secondary prevention. Mnemonic: 'after MI, aspirin (± colchicine) — steroids and NSAIDs spoil the scar.'

BOARD TRAP

Board says: 'Give corticosteroids (or indomethacin) for post-MI pericarditis.' WRONG. Discriminator: steroids and non-ASA NSAIDs impair myocardial healing and increase rupture risk early post-MI — the correct treatment is aspirin (± colchicine).

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13. LV aneurysm comparison

WHAT YOU SEE

Bottom display 'ANEURYSM COMPARISON' showing two LV jars side by side: a WIDE-MOUTHED smooth bulge (true aneurysm) versus a NARROW-NECKED, pericardium-wrapped jar (pseudoaneurysm) — the neck width is the whole distinction.

WHAT IT ENCODES

Distinguish a TRUE LV aneurysm from a PSEUDOaneurysm. True aneurysm: a wide-mouthed, smooth bulge of all myocardial layers (scarred but intact wall), forms weeks after MI, low rupture risk; complications are HF, persistent ST elevation on ECG, mural thrombus, and refractory VT. Pseudoaneurysm: a contained free wall rupture walled off by pericardium/thrombus, with a NARROW neck and HIGH rupture risk → surgical emergency. Mnemonic: 'true = wide & tame; pseudo = narrow neck & nasty.'

BOARD TRAP

Board says: 'A true LV aneurysm ruptures more often than a pseudoaneurysm.' WRONG. Discriminator: the PSEUDOaneurysm (narrow neck, only pericardium containing the rupture) has the HIGH rupture risk and needs urgent surgery; a true aneurysm rarely ruptures.

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14. AIVR exhibit

WHAT YOU SEE

Bottom-right 'AIVR EXHIBIT': a calm man strolling 'at a leisurely pace' beside '60–100 BPM' and a 'DON'T TREAT THIS' tag with amiodarone X'd out — the relaxed slow walk = the benign reperfusion rhythm you leave alone.

WHAT IT ENCODES

Accelerated idioventricular rhythm (AIVR) is a wide-complex ventricular rhythm at a rate of ~40–100/min (slow VT-like) that is a classic, BENIGN marker of successful REPERFUSION after thrombolysis or PCI. It is hemodynamically stable and self-limited; suppressing it (e.g., with lidocaine) is harmful because it may be the patient's only escape rhythm. Mnemonic: 'AIVR = reperfusion's calling card — leave it alone.'

BOARD TRAP

Board says: 'AIVR after reperfusion requires urgent antiarrhythmic suppression with lidocaine/amiodarone.' WRONG. Discriminator: AIVR is benign and self-resolving — the correct answer is observation ('don't treat'); antiarrhythmics can abolish a needed escape rhythm and cause asystole.
